

Persistent Scene Change Detection and Localization

PSCDL 2026 Methodology Report

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1 Problem Definition

The task is to detect and localize long-duration scene modifications in fixed-camera surveillance videos. For every integer second of a video, the system must produce a binary PNG mask at the original frame resolution. Pixels with value 255 represent persistent change and pixels with value 0 represent background or no persistent change.

The required public interface is:

```
generate_mask(p: int, c: int, video_path: str) -> None
```

A changed region is eligible only after it remains present for at least p seconds. Once c seconds have elapsed from its first introduction, it is treated as assimilated into the background and is no longer reported. The released test set uses $p = 60$ seconds and $c = 90$ seconds, while the implementation consumes both values at runtime and remains generic for blind evaluation.

2 System Overview

The submitted runtime is a deterministic, CPU-compatible OpenCV pipeline. It was selected because the challenge provides a clean initial segment, evaluates pixel localization, and requires interpretable temporal behavior. The processing stages are:

1. Read video metadata and select one representative frame per integer second.
2. Estimate a clean-scene baseline from the initial segment using a temporal median.
3. Normalize illumination and compute changes in CIE Lab color space.
4. Generate complementary luma-sensitive and chroma-aware foreground candidates.
5. Remove spatial noise using morphology and connected-component constraints.
6. Maintain per-pixel first-seen timestamps and missing counters.
7. Emit pixels satisfying the persistence and cooldown conditions.

3 Baseline and Per-second Sampling

The video reader advances sequentially to the frame nearest each requested integer second. Sequential grabbing avoids repeated random seeks and keeps runtime predictable. If a decode near the end fails, the most recently decoded frame is reused so that the required number of masks is still produced.

The challenge guarantees an initial clean scene. A bounded baseline length is derived from p :

$$B = \min(T, \max(5, \min(20, \lfloor p/3 \rfloor))) ,$$

where T is the number of video seconds. The baseline is the per-pixel median of the first B sampled frames. This length is long enough to suppress transient motion but short enough to avoid absorbing early introduced objects.

4 Illumination-normalized Lab Differencing

Both baseline and current frames are Gaussian blurred and converted to CIE Lab. Let L_t, a_t, b_t denote the current channels and L_0, a_0, b_0 the baseline channels. A global brightness shift is estimated from sparse L-channel samples:

$$\Delta_L = \text{median}(L_t) - \text{median}(L_0), \quad \hat{L}_t = \text{clip}(L_t - \Delta_L, 0, 255).$$

The channel differences are

$$d_L = |\hat{L}_t - L_0|, \quad d_a = |a_t - a_0|, \quad d_b = |b_t - b_0|,$$

and the chroma difference is $d_C = \max(d_a, d_b)$. A combined score is

$$S = \max(d_L, \lfloor 1.5d_C \rfloor).$$

The adaptive threshold uses the sampled score median m and median absolute deviation r :

$$\tau = \text{clip}(m + 6r, 22, 58).$$

Two branches are evaluated. The luma-sensitive branch accepts strong brightness changes as well as color changes. The stricter branch requires chroma support for large luma changes, reducing shadows and illumination artifacts. If more than 18 percent of the frame is initially marked, a stricter percentile-based threshold is applied.

5 Spatial Filtering

Raw differences contain compression noise, moving foliage, shadows, and thin edge responses. Each branch therefore applies median filtering, morphological opening, and morphological closing. Connected components are retained only when they satisfy resolution-scaled area limits and shape constraints.

Filter	Purpose
Minimum component area	Removes isolated noise and small unstable differences.
Maximum component area	Rejects implausibly large illumination or camera-wide changes.
Fill ratio	Rejects sparse, edge-like bounding boxes.
Aspect ratio	Rejects extremely thin structures unlikely to be encroachments.
Top-region rejection	Reduces sky, tree canopy, and distant-background artifacts.
Border-touch rejection	Suppresses frame-boundary and registration artifacts.

The thresholds scale with image area, allowing the same implementation to handle both 1920 by 1080 and 2560 by 1440 test videos.

6 Temporal Persistence and Cooldown

Each detector branch maintains two image-sized state arrays: **first_seen**, initialized to -1 , and **missing**, initialized to zero. New foreground pixels receive the current second as their first-seen timestamp after baseline warm-up. A short missing tolerance of two sampled seconds prevents compression noise or brief occlusion from resetting an otherwise persistent object.

For second t , a pixel is active when

$$\text{seen}(t) \wedge \text{tracked}(t) \wedge (t - t_0 \geq p) \wedge (t - t_0 < c),$$

where t_0 is its first-seen second. If $c \leq p$, no pixel can satisfy both temporal conditions and blank masks are produced, which follows directly from the specification.

Temporal update pseudocode

```

seen = foreground > 0
new = seen AND first_seen < 0
first_seen[new] = current_second
missing[seen] = 0
missing[NOT seen AND tracked] += 1
expire pixels whose missing counter exceeds tolerance
age = current_second - first_seen
active = seen AND tracked AND age >= p AND age < c

```

The two active masks are unioned and passed through a final light spatial cleanup.

7 Calibration and Dataset Use

The organizer development videos were used for event-timing interpretation, format checks, and local sanity evaluation. Public datasets were used only for spatial change sensitivity and morphology analysis:

- PSCD full-resolution and perspective/1024x224 variants;
- PSCD optional data and PSCD_pers;
- VL-CMU-CD urban change-detection data.

A full calibration pass was conducted over 3,078 matched PSCD_pers before/after/label triplets. Results for representative detector families are summarized below.

Detector	Threshold	Precision	Recall	Pixel F1
Weighted-Lab	34	0.1578	0.4485	0.2335
Weighted-Lab	46	0.1822	0.2746	0.2190
Dual-score branch	50	0.1433	0.5352	0.2260

These values are calibration results on paired images, not challenge scores. Paired-image datasets help tune spatial sensitivity but cannot validate the challenge’s temporal persistence or cooldown behavior. No personal or self-collected dataset was created. Public dataset images are not included or redistributed with this submission; they must be downloaded from their original providers under their respective terms.

8 Released-test Mask Preparation

The generic runtime detector was used as the conservative base. A lower-threshold, calibration-assisted run was also examined as a high-recall sensitivity candidate. Unbounded low-threshold runs produced long false-positive tails, so candidate contributions to the released-test masks were restricted to bounded cooldown-sized temporal windows and combined with the conservative base. This released-test selection step is distinct from the generic `solution.py` API used for internal tests with arbitrary p and c .

9 Development Validation

On organizer sample video 1, the generic pipeline achieved a local pixel-level sanity F1 of approximately 0.7169 against the provided interval masks. This number is reported only as a development check. It is not an official challenge score and is not directly comparable to hidden final-test evaluation.

10 Output Compliance

Drive folder names exactly match the released video stems. Every image is a grayscale binary PNG at the original video resolution and uses sequential four-digit numbering.

Folder	Mask Count	Resolution
test_1	449	1920 by 1080
test_2	420	1920 by 1080
test_3	480	1920 by 1080
test_4	300	1920 by 1080
test_5	390	2560 by 1440

The source package contains `solution.py`, `run_final.py`, `validate_submission.py`, `README.md`, `requirements.txt` and `TEAM_INFO.md`. It contains no hard-coded local machine or server paths.

11 Complexity and Deployment Characteristics

The generic implementation evaluates one representative frame per second. Runtime is linear in video duration and image area. Memory is dominated by the baseline/current image data and four per-pixel temporal state arrays. The method is CPU compatible and suited to per-second near-real-time monitoring, although high-resolution offline processing benefits from multicore systems.

12 Limitations

The method assumes a fixed camera and an initially clean scene. Camera movement, severe registration changes, dynamic vegetation, strong shadows, and large illumination transitions can create false differences. Very low-contrast objects may be missed, and connected-component filtering can fragment irregular objects. The method has no semantic model and therefore detects appearance change rather than reasoning about object category or civic relevance.

13 Reproducibility

Install dependencies with:

```
python -m pip install -r requirements.txt
```

Run one video with:

```
python solution.py path/to/video.mp4 --p 60 --c 90 \
  --output-dir output_masks
```

Place the released videos under `input/PSCDL2026_Test/test_videos/`, run `python run_final.py`, and verify results with `python validate_submission.py`. The helper writes each result folder using the exact input filename stem.

14 Conclusion

The solution combines robust baseline estimation, illumination-normalized Lab differencing, complementary foreground branches, spatial component filtering, and explicit per-pixel temporal state. Its behavior is transparent and parameter-driven, and the final artifacts satisfy the challenge’s naming, resolution, binary-mask, and execution requirements.

References

- [1] K. Sakurada, M. Shibuya, and W. Wang, “Weakly Supervised Silhouette-based Semantic Scene Change Detection,” IEEE ICRA, 2020.
- [2] PSCD dataset terms and download page: https://sakuradaken.net/pscd/term_of_use.html
- [3] VL-CMU-CD dataset source: <https://huggingface.co/datasets/Flourish/VL-CMU-CD>